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A Kresling-Based Origami Electromagnetic Energy Harvester

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Abstract. Energy harvesting using smart structures has attracted significant attention due to its wide range of applications, including the Internet of Things (IoT), microelectromechanical systems (MEMS), data acquisition, biological systems, offshore technologies, and civil infrastructure. In parallel, origami-inspired designs have driven advances in engineering by enabling the development of novel adaptive and morphing structures. Among these, the Kresling origami pattern has emerged as a promising concept due to its intrinsic coupling between rotational and axial motions, its lightweight architecture in and capacity to achieving bistability, which together provide enhanced mechanical performance. In this regard bistable systems are known to improve energy harvesting efficiency by enabling large-amplitude responses and nonlinear dynamics. In this work, a novel Kresling-based energy harvesting system is proposed. An electromagnetic transduction mechanism is used to convert mechanical to electrical energy, taking advantage of its compatibility with the Kresling geometry. Two configurations are selected: a monostable and a bistable. Both designs exhibit a maximum average output power of approximately 0.5 W, while the bistable configuration manifests a larger operational bandwidth.
Keywords: Energy Harvesting, Origami, Kresling, Nonlinear Dynamics, Electromagnetic

INTRODUCTION

Energy harvesting technologies offer a promising approach for enhancing smart cities capabilities, expanding the integrated use of renewable energy, and accelerating the development of sustainable infrastructure. In addition, these devices can be applied in Internet of Things (IoT), microelectromechanical systems (MEMS), data acquisition, and biomedical systems. Consequently, energy harvesters can be integrated into roadways, subsea pipelines, and even wearable technologies. One common approach consists of converting mechanical vibration into electrical energy through electromechanical transducers, such as piezoelectric, magnetostrictive, triboelectric, and electromagnetic transducers.

Among the various strategies for improving the performance of vibration energy harvesters, multistability has attracted considerable attention due to its ability to broaden the operational bandwidth and increase the amount of harvested energy under broadband or low-frequency excitations. By enabling large-amplitude interwell motions, multistable systems can significantly outperform their linear counterparts, making them an attractive solution for ambient vibration energy harvesting (Costa *et al.*, 2021; Costa and Savi, 2024).

Origami-inspired structures have emerged as promising candidates for multistable energy harvesting, as they are lightweight, compact, deployable, and reconfigurable. Their geometric nonlinearities naturally give rise to multiple stable configurations, making them particularly suitable for broadband vibration energy harvesting. Previous studies have demonstrated that origami-based systems can harvest energy from wind, assist solar-driven water purification processes, and convert vibratory motion from ocean waves into electrical energy (Fonseca *et al.*, 2022).

The Kresling origami is a flat-foldable, multistable structure that exhibits axial-twist coupling and has been studied for the development of energy harvesting devices. Chung *et al.* (2021) proposed an origami-based piezoelectric/triboelectric hybrid generator, leveraging the Kresling origami axial-twist coupling to harvest both axial and rotational motions. In their design, triboelectric nanogenerators (TENG) were installed on the origami panels and on the top substrate, while piezoelectric nanogenerators (PENG) were embedded along the inner hinges.

Huang *et al.* (2022a,b) developed a Kresling origami energy harvester with piezoelectric films attached to the mountain creases. Their dynamic model does not account for the Kresling origami mass, as it is negligible compared with the mass of the block on top of the structure. More recently, Yin *et al.* (2024) proposed a Kresling origami electromagnetic energy harvester consisting of a coil mounted at the origami top and a permanent magnet attached to a plate above it. Their prototype achieved an RMS output voltage ranging from 0.09 to 0.20 V and a maximum output power of approximately 0.5 W.

Electromechanical coupling has also been exploited by Kortman *et al.* (2025), in which a Kresling origami was set between two electromagnets composed of a multilayer solenoidal coils with ferromagnetic cores.

Inspired by the design proposed by Kortman *et al.* (2025), this work presents a novel Kresling origami electromagnetic energy harvester based on an electrical coil and a cylindrical permanent magnet. First, the proposed device is introduced, highlighting how its compactness is achieved by embedding the magnet inside the origami structure. Next, a coupled electromechanical model is derived using an energy-based formulation, and the mechanical behavior of the Kresling origami unit cell is analyzed to characterize its multistable response. Finally, the dynamic behavior and energy harvesting performance of both monostable and bistable configurations are investigated to demonstrate the feasibility and effectiveness of the proposed harvester.

DESIGN AND MODELING

This section presents the design concept of the proposed Kresling-based origami electromagnetic energy harvester and outlines the assumptions adopted for its mathematical modeling.

A schematic representation of the proposed device is provided in Fig. 1. The device consists of a Kresling origami structure whose base is rigidly attached to a cylindrical support. Additionally, a hollow cylindrical proof mass, featuring an open-box-shaped cross-section, is connected to the free end of the origami. An electrical coil is wound around the outer surface of the proof mass. Moreover, a cylindrical permanent magnet is fixed at the top center of the supporting base and aligned with the coil's center. The geometry is designed so that the coil can move freely around the magnet as the Kresling structure undergoes axial displacement and torsion during folding and unfolding.

When the base is subjected to external mechanical excitation, the proof mass oscillates relative to the fixed magnet, causing a time-varying magnetic flux through the coil. According to Faraday's law of electromagnetic induction, this varying flux induces a voltage across the coil terminals.

Therefore, by connecting the coil to an external circuit and leveraging Kresling origami features such as multistability, shape-changing capability, and axial-twist coupling, this conceptual design provides a novel approach to efficiently harvest the induced electrical energy for practical use.

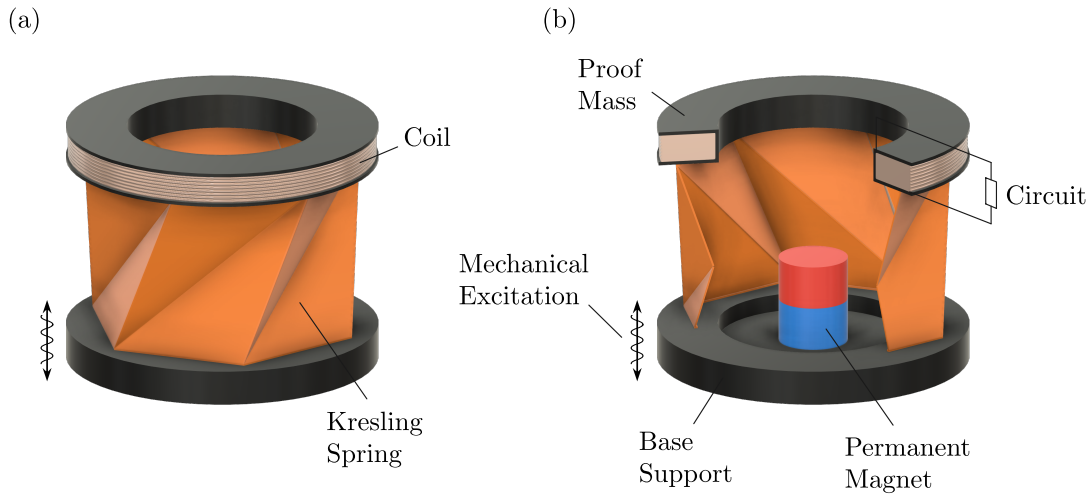


Figure 1 – Conceptual representation of the Kresling-based origami electromagnetic energy harvester: (a) external view and (b) cylindrical cross-sectional view showing internal components.

The energy harvester presented in Fig. 1 can be represented by a general lumped model as depicted in Fig. 2. The mechanical characteristics of the system are represented by the effective mass, m, the mechanical dissipation factor, γ, and the Kresling nonlinear restitution force, $f(u)$, with $u(t)$ being the generalized coordinate that describes the relative displacement between the proof mass and the base support. Additionally, the electrical characteristics of the system are represented by electrical resistance, R, and the coil self-inductance, $L = 31.5 r_m^2 N_c^2/[6r_m + 9h_c + 10(r_2 - r_1)] \times 10^{-6}$ H, where $r_m = (r_1 + r_2)/2$ is the mean radius of the coil (Wheeler, 1928). The electrical charge, $q(t)$, is the generalized coordinate of the electrical domain, with the electrical current being defined as the temporal derivative of the charge $I(t) = \dot{q}(t)$.

The coupling between the mechanical and electrical domains is represented by $\vartheta(u)$, which depends on the geometrical characteristics of the coil and the magnetic and geometrical properties of the central magnet. r_1, r_2, and h_c are the internal radius, the external radius, and the height of the coil, respectively. The maximum height of the Kresling origami nonlinear

spring is denoted as h_0, and the distance between the center of the magnet and the inertial reference frame is denoted as h_m. Additionally, $z_1(t) = h_0 + h_m - u(t)$ and $z_2(t) = z_1(t) + h_c$ are the distances between the coil vertical ends and the center of the magnet. Moreover, the system is subjected to a base excitation of $u_b = A\sin(\omega t)$, where A and ω represent the excitation amplitude and frequency, respectively.

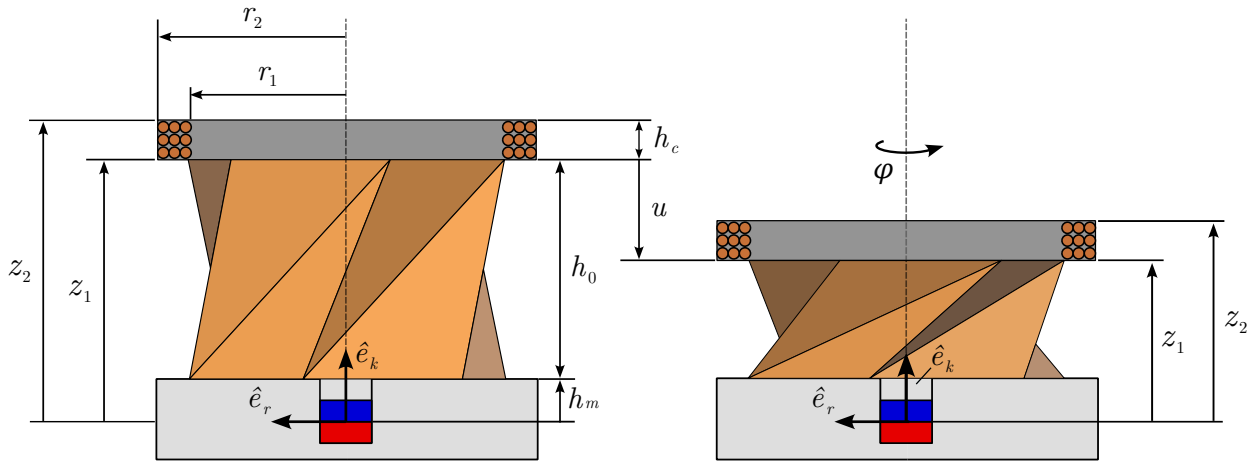


Figure 2 – Archetype representing the Kresling-based origami electromagnetic energy harvester.

An energetic approach is carried out to determine the electromechanical equations of the energy harvester. The Euler-Lagrange equations are defined from 1 mechanical and 1 electrical generalized coordinates, $\mathbf{x} = [u(t), q(t)]$, such that

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}_i}\right) - \frac{\partial \mathcal{L}}{\partial x_i} + \frac{\partial \mathcal{D}}{\partial \dot{x}_i} = 0, \tag{1}$$

where the Lagrangian, $\mathcal{L} = T - U + W_{em}$, is given by the sum of the kinetic energy, T, the potential energy U, and the total electromagnetic energy, W_{em}. Furthermore, the total dissipation is defined from the electromechanical dissipation function $\mathcal{D}$.

Equation (2) represents the kinetic energy of the system. In Eq. (3), the charge formulation is used to determine the total energy stored in the electromagnetic transducer, W_{em}, which is composed of the sum of the magnetic energy in the inductance, W_L, and the electromechanical coupling energy, W_ϑ (Preumont, 2006).

$$T = \frac{1}{2} m \left[\dot{u}(t) + \dot{u}_b(t)\right]^2, \tag{2}$$

$$W_{em} = W_L + W_\vartheta = \frac{1}{2} L \dot{q}(t)^2 + \dot{q}(t) \int_0^{u(t)} \vartheta(u)\, du, \tag{3}$$

where the electromechanical coupling is defined by Eq. (4) based on the assumption that the magnetic field is considered a point dipole and the coil is finite and short (Owens and Mann, 2012).

$$\vartheta(u) = \frac{\xi N_c B_r V_m}{2A_c} \left\{ \sum_{i=1}^{2} \sum_{j=1}^{2} (-1)^{i+j} \left[\ln\left(r_i + \Gamma_{ij}\right) - \frac{r_i}{\Gamma_{ij}} \right] \right\}. \tag{4}$$

Here, A_c denotes the coil cross-sectional area, and ξ is the fill factor, which represents the fraction of the coil volume effectively occupied by conductive wire. The total number of coil turns is given by $N_c = \xi A_c / A_w$, where A_w is the cross-sectional area of the conductor wire. The parameters B_r and V_m characterize the permanent magnet, with B_r denoting the remanent magnetic flux density and V_m its volume. $\Gamma_{ij} = \sqrt{r_i^2 + z_j^2}$ is the absolute value of the distance of the magnet to a point defined by the coordinates r_i and z_j.

The dissipation is mainly composed by structural viscous forces, internal magnetic dissipation within the magnetic transducer, and resistive circuit elements, that can be expressed as:

$$\mathcal{D} = \mathcal{D}_s + \mathcal{D}_m + \mathcal{D}_R = \frac{1}{2} c_s \dot{u}(t)^2 + \frac{1}{2} c_m \dot{u}(t)^2 + \frac{1}{2} R \dot{q}(t)^2 \tag{5}$$

The Kresling origami cell exhibits coupled axial–torsional motion, thus producing simultaneous axial displacement and twisting deformation. This unique characteristic makes it particularly attractive for energy harvesting applications, as its potential energy is predominantly associated with the elastic energy stored in the Kresling spring.

The amount of elastic energy stored during deformation is governed by the origami geometry and the mechanical properties of the constituent material. The geometry of a Kresling cell is defined by a set of parameters, illustrated in Figure 3, including the folded height, h_f, the initial relative rotation angle between the top and bottom polygons or design angle, θ_0, and the radius of the circumscribed circle of the polygon, r.

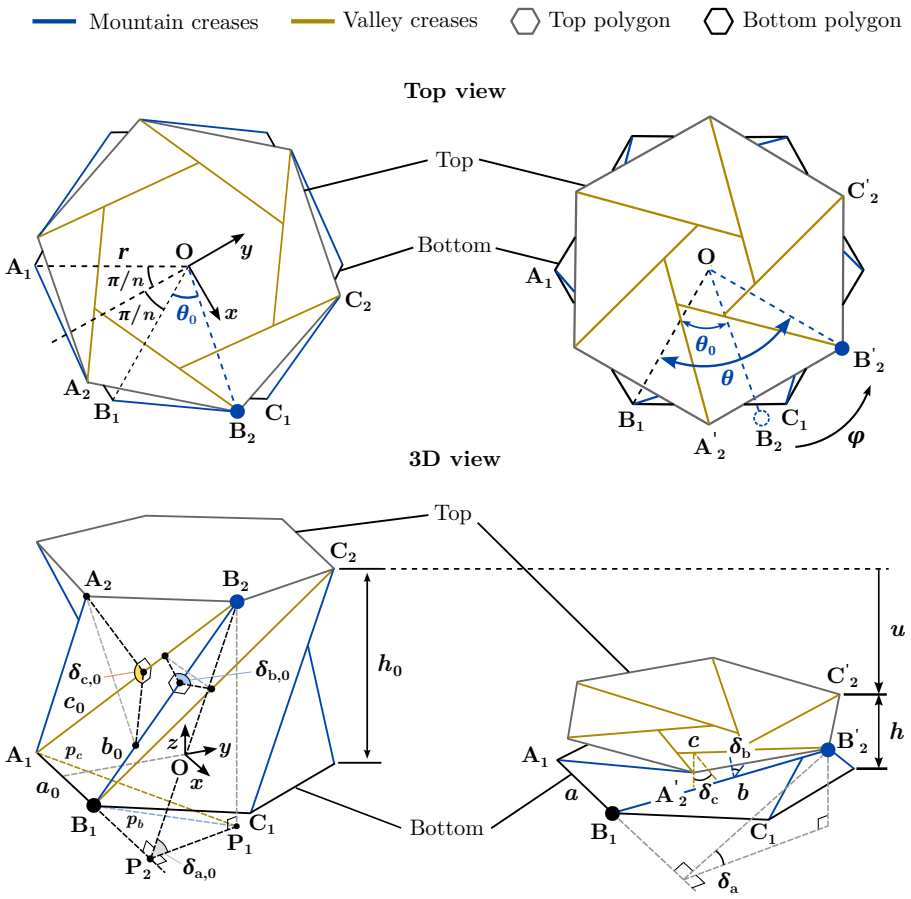


Figure 3 – Kresling origami cell top view and 3-D view. Adapted from Zang *et al.* (2024).

The lengths a, b, and c, and the dihedral angles δ_a, δ_b, and δ_c are described by

$$a = 2r\sin(\pi/n) \tag{6}$$

$$b(u,\varphi) = \sqrt{h^2 + 4r^2\sin^2(\theta/2)} \tag{7}$$

$$c(u,\varphi) = \sqrt{h^2 + 4r^2\sin^2(\theta/2 + \pi/n)} \tag{8}$$

$$\delta_a(u,\varphi) = \tan^{-1}\left\{\frac{h}{2r\sin(\theta/2+\pi/n)\sin(\theta/2)}\right\} \tag{9}$$

$$\delta_b(u,\varphi) = \pi - \cos^{-1}\left\{\frac{h^2\cos(\theta+2\pi/n) - r^2[\cos(\theta+\pi/n) - \cos(\pi/n)]^2}{h^2 + r^2[\cos(\theta+\pi/n) - \cos(\pi/n)]^2}\right\} \tag{10}$$

$$\delta_c(u,\varphi) = \pi - \cos^{-1}\left\{\frac{h^2\cos(\theta) - r^2[\cos(\theta+\pi/n) - \cos(\pi/n)]^2}{h^2 + r^2[\cos(\theta+\pi/n) - \cos(\pi/n)]^2}\right\} \tag{11}$$

where $h = h_0 - u, u \in [h_f, h_0]$, is the height of the Kresling origami cell, and $\theta = \varphi + \theta_0, \varphi \in [0, \theta_{\max} - \theta_0]$, is the twisting angle of the top polygon with bottom polygon fixed, u is the distance between points B_2 and B_2' in the z-direction, and φ is the twisting angle of the top polygon with the bottom polygon fixed.

Based on the Kresling geometrical characteristics, the total elastic energy, $U(u,\varphi)$, is obtained considering the bar-and-hinge formulation from Liu and Paulino (2017) along with the 5 parameter model proposed by Zang *et al.* (2024):

$$\begin{aligned} U(u,\varphi) &= U_{\text{bar}}(u,\varphi) + U_{\text{rot}}(u,\varphi) \\ &= \frac{1}{2}\Big\{ n_b k_{s,b} \left[b(u,\varphi) - b_0\right]^2 + n_c k_{s,c} \left[c(u,\varphi) - c_0\right]^2 \\ &\quad + n_a k_{r,a} \left[\delta_a(u,\varphi) - \delta_{a,0}\right]^2 + n_b k_{r,b} \left[\delta_b(u,\varphi) - \delta_{b,0}\right]^2 \\ &\quad + n_c k_{r,c} \left[\delta_c(u,\varphi) - \delta_{c,0}\right]^2 \Big\} \end{aligned} \tag{12}$$

where U_{bar} is the elastic energy stored in bar elements, U_{rot} is the elastic energy of the rotational spring elements, n_a is the number of edges of top and bottom surfaces, n_b is number of mountain creases, n_c is the number of valley creases, b is the length of mountain creases, b_0 is the initial length of mountain creases, c is the length of valley creases, c_0 is the initial length of valley creases, δ_a is the dihedral angle between surface polygons and side triangular panels, $\delta_{a,0}$ is the initial dihedral angle between surface polygons and side triangular panels, δ_b is the dihedral angle between side triangular panels along mountain folds, $\delta_{b,0}$ is the initial dihedral angle between side triangular panels along mountain folds, δ_c is the dihedral angle between side triangular panels along valley folds, $\delta_{c,0}$ is the initial dihedral angle between side triangular panels along valley folds, $k_{s,b}$ is the stretching stiffness of mountain creases, $k_{s,c}$ is the stretching stiffness of valley creases, $k_{r,a}$ is the folding stiffness between surface polygons and side triangular panels, $k_{r,b}$ is the folding stiffness between side triangular panels along mountain folds, and $k_{r,c}$ is the folding stiffness between side triangular panels along valley folds.

On this basis, the electromechanical equations related to the state variable, $u(t)$ and $I(t)$, are found by substituting Eqs. (2), (3), (5), and (12) into (1):

$$m\ddot{u}(t) + \gamma\dot{u}(t) + \frac{\partial U(u,\varphi)}{\partial u} - \vartheta(u) I(t) = -m\ddot{u}_b, \tag{13}$$

$$L\dot{I}(t) + RI + \vartheta(u)\dot{u}(t) = 0, \tag{14}$$

If the gravitational effect of the hollow cylindrical proof mass weight is taken into account, a shift in the Kresling origami equilibrium point at its deployed configuration is calculated as

$$\frac{\partial U(u,\varphi)}{\partial u}\Big|_{u=\tilde{u}} = mg \tag{15}$$

where $\tilde{u}$ is the shifted equilibrium point due to gravity, g. In this work, gravity effects are neglected.

Table 1 provides the parameters of the Kresling origami electromagnetic energy harvester. A conservative coil fill factor of $\xi = 0.7$ is used, based on the values provided in McLyman (2004).

Performance Metrics

The performance of the proposed energy harvester is assessed through its electrical output characteristics. Both instantaneous and time-averaged quantities are considered to provide a comprehensive evaluation of the device. To characterize the raw electrical output, the harvester is connected to the simplest load configuration, namely a purely resistive circuit. In this case, the instantaneous electrical power dissipated across the load resistor is given by Eq. (16):

$$P(t) = RI(t)^2. \tag{16}$$

The average output power is obtained by integrating the instantaneous power over the evaluation interval and dividing by its duration, as expressed in Eq. (17):

$$P_{\text{avg}} = \frac{1}{t_f - t_0}\int_{t_0}^{t_f} P(t)\,dt = RI^{\text{RMS}}, \tag{17}$$

where t_0 and t_f denote the initial and final times of the evaluation interval, respectively, and I_{RMS} is the root-mean-square (RMS) current, defined as

$$I_{\text{RMS}} = \sqrt{\frac{1}{t_f - t_0}\int_{t_0}^{t_f} I(t)^2\,dt} \tag{18}$$

Table 1 – Parameters of the Kresling origami electromagnetic energy harvester.

Parameter	Description	Value	Unit
$h_{0,M}$	Monostable Kresling origami initial height	20	mm
$h_{0,B}$	Bistable Kresling origami initial height	28	mm
r	Radius of the circumscribed circle of the polygon	20	mm
n	Number of edges of the polygon	6	-
$\theta_{0,M}$	Design angle of the monostable Kresling origami	32	deg
$\theta_{0,B}$	Design angle of the bistable Kresling origami	30	deg
N_c	Number of coil turns	1658	-
L	Coil self-inductance	142739	μH
R	Coil Electrical Resistance	262.46	Ω
$\Delta\vartheta_M$	Electromechanical coupling coefficient range - monostable	1994.08	N/A
$\Delta\vartheta_B$	Electromechanical coupling coefficient range - bistable	2092.42	N/A
r_1	Inner radius of the coil	16	mm
r_2	Outer radius of the coil	22	mm
h_c	Height of the coil	5	mm
m	Top surface mass	45.88	g
h_m	Inertial reference frame	3	mm
r_m	Magnet radius	8	mm
h_{magnet}	Magnet height	6	mm
m_{magnet}	Magnet mass	9.05	g
V_m	Magnet volume	1206.37	mm^3
B_r	Remanent magnetic flux density	1.29	T

NUMERICAL SIMULATIONS

This section presents numerical simulations of the proposed model to characterize the system and evaluate its energy harvesting performance.

Geometric and Electromechanical Assessment

First, a stability analysis is performed based on the geometric characteristics of the Kresling origami. One monostable and one bistable configuration are selected to investigate the energy harvesting performance. The geometries of both configurations are designed so that their peak restoring forces occur at similar magnitudes, enabling a fair comparison between the two stability regimes. Next, the electromechanical coupling coefficient is evaluated for both stability configurations.

The main geometric parameters of the Kresling origami are listed in Table 1. The total elastic energy profile and the electromechanical coupling coefficient, shown in Fig. 4, are obtained from Eqs. 12 and 4, respectively. For the monostable configuration, the equilibrium position is located at $u_I = u_1 + \tilde{u} = 1.27$ mm. For the bistable configuration, the two equilibrium positions are located at $u_I = u_1 + \tilde{u} = 1.26$ mm and $u_{II} = u_2 + \tilde{u} = 18.20$ mm.

Performance Evaluation

Based on the two selected configurations, a performance evaluation is carried out under different excitation conditions. Specifically, the system performance is investigated in the excitation amplitude–frequency parameter space, A (mm) $\times$ ω (Hz). The excitation frequency varies from 10 to 300 Hz to cover typical environmental excitations, and the amplitude ranges from from $h_0/4$ to h_0 relative to the structure's height. An Output Power Diagram (OPD) is constructed, where each point corresponds to a numerical solution of the governing equations obtained using the fourth-order Runge–Kutta method. A time step of $dt \propto 2\pi/\omega$, proportional to the excitation period, is adopted based on a convergence analysis. The initial conditions are set as $u(0) = \dot{u}(0) = I(0) = 0$ for every simulation. To improve computational efficiency, a polynomial approximation is employed to express the total elastic energy, U, exclusively as a function of the axial displacement, u. This approach significantly reduces the computational cost while accurately preserving the nonlinear restoring behavior of the Kresling origami. The corresponding results are presented in Fig. 5.

A maximum average output power in the order of 0.5 W is achieved by the energy harvester. Furthermore, comparing

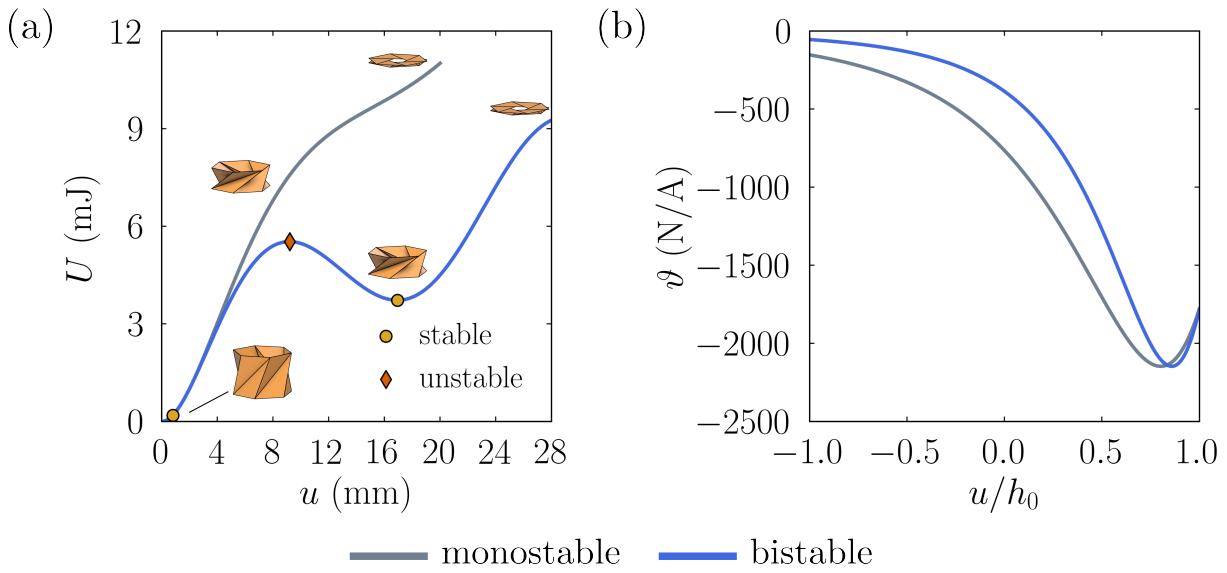


Figure 4 – Kresling origami mechanical behavior. (a) Monostable behavior with equilibrium point located at $u_I = 1.27$ mm, and Bistable behavior with the two equilibrium points located at $u_I = 1.26$ mm and $u_{II} = 18.20$ mm. (b) Electromechanical coupling coefficient for the monostable and bistable Kresling origami energy harvester.

the performance of the two configurations across the excitation parameter space reveals that the bistable configuration exhibits a broader operational bandwidth, resulting in enhanced energy harvesting performance over a wider range of excitation frequencies, particularly at higher frequencies.

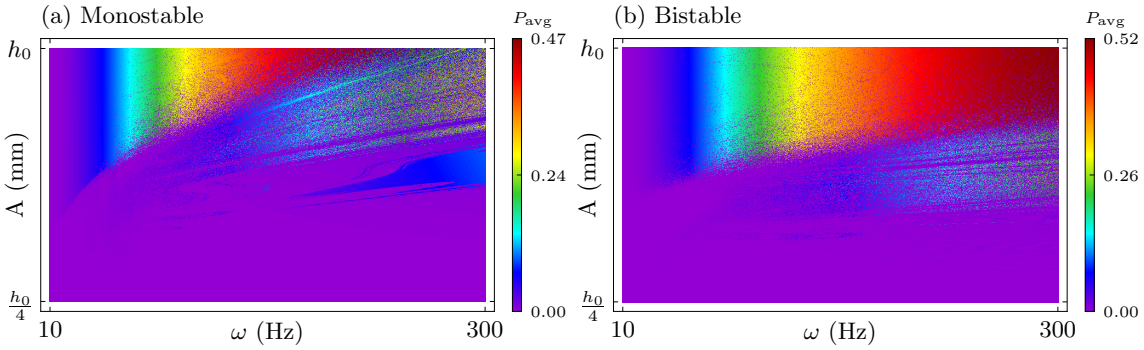


Figure 5 – Kresling origami performance in the operation parameter space: (a) for the selected monostable case; (b) for the selected bistable case. Color bar represents levels of average output power, P_{avg}, in Watts [W].

CONCLUSION

This work presents a novel electromagnetic energy harvester based on a Kresling origami structure, employing an electrical coil and a cylindrical permanent magnet. A conceptual design is introduced to highlight the proposed harvesting mechanism, and an analytical electromechanical model is developed to describe the coupled dynamics through the state variables $u(t)$ and $I(t)$. In addition, a geometric analysis of the Kresling origami is performed to identify the parameters h_0, r, and θ_0 that produce monostable and bistable behaviors. The dynamic response and energy harvesting performance are then evaluated for representative monostable and bistable configurations.

The numerical results demonstrate that the proposed energy harvester can achieve a maximum average output power of in the order of 0.5 W. Furthermore, the bistable configuration exhibits a broader operational bandwidth and superior energy harvesting performance over a wider range of excitation conditions, particularly at higher excitation frequencies, when compared with the monostable configuration. These findings highlight the potential advantages of exploiting multistability

to enhance the performance and robustness of origami-based electromagnetic energy harvesters.

Future work will focus on investigating additional geometric configurations of the proposed concept, optimizing the electromechanical design, and fabricating a prototype to experimentally validate the analytical and numerical predictions.

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